

Vehicle Mileage & Reservation Tracker

Excel/VBA operational application | Independently designed and built end to end by Charles Monsanto

BUSINESS PROBLEM

Small fleets are often coordinated across disconnected spreadsheets, calendars, and messages. That fragmentation makes booking conflicts, dated vehicle status, controlled corrections, and consistent reporting difficult to manage.

DELIVERED SOLUTION

- One controlled desktop workflow for mileage, reservations, maintenance history, and management reporting.
- Pre-write validation for required fields, dates, times, vehicles, drivers, odometers, conflicts, and maintenance state.
- Recent-entry edit and delete workflows that recalculate dependent odometers, metrics, and reports.
- Protected generated sheets with approved dashboard actions preserved for routine use.

VEHICLE MILEAGE AND RESERVATION TRACKER

LOG MILEAGE

Trip Date

7/20/2026

CAL

Vehicle

Odo Start

Odo End

SAVE MILEAGE

CLEAR

GO TO MILEAGE TRACKER TAB

RESERVE VEHICLE

Date

7/20/2026

CAL

Vehicle

Staff

Start

8:00 AM

End

5:00 PM

Purpose

SAVE RESERVATION

CLEAR

GO TO RESERVATIONS TAB

SHOP STATUS

Vehicle

Location

Shop - Central Garage

MARK IN SHOP

MARK AVAILABLE

RESERVATIONS

Open Reservations tab to manage weekly bookings.

GO TO RESERVATIONS TAB

STATUS LEGEND:

Reserved

Available

In Shop

MONTHLY MILEAGE REVIEW

Fiscal Year

FY 2026-27

Report Month

July 2026

Vehicle	Last Mileage	Checkout Status (This Week)	Month Miles	Trip Days	Trips	Scheduled Hours	FYTD Miles
Demo SUV	52,530	Available	530	5	5	16.0	530
Demo Crossover	41,925	Shop - Central Garage	225	2	2	0.0	225
Demo Compact	28,740	7/20 Reserved - Cameron Davis	440	4	4	32.0	440
Demo Pickup	63,110	Available	310	3	3	16.0	310
TOTAL			1,505	14	14	64.0	1,505

Note: Please ensure all mileage and checkout information is accurate before saving.

CLEAR FIELDS

EDIT MILEAGE

EDIT RESERVATION

Synthetic dashboard view: entry, maintenance, correction, and reporting controls.

INDEPENDENT OWNERSHIP

Charles Monsanto owned requirements discovery, workflow and control design, VBA implementation, reporting, test design, privacy sanitization, and public-release documentation. The result is a reviewable application, not a presentation-only prototype.

LIGHTWEIGHT ARCHITECTURE

DASHBOARD

Constrained inputs

>

VBA CONTROLS

Validation + rules

>

EXCEL TABLES

Local structured data

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OUTPUTS

Reports + metrics

Structured Excel ListObject tables provide the local data store. Thin worksheet event modules dispatch actions to three standard VBA modules. No server, external database, add-in, API, or data connection is required.

VERIFIED TECHNICAL PERFORMANCE

1.844 s

FUNCTIONAL OPEN

Measured on the test machine

0.219 s

DASHBOARD REFRESH

Measured on the test machine

Vehicle Tracker | Portfolio Overview

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Workflow Integrity by Design

Reservation rules, correction controls, maintenance history, reporting, and release verification

WEEKLY RESERVATIONS

Select week

Month

July

Year

2026

<

Month

>

Date

7/20/2026

CAL

PREVIOUS WEEK

NEXT WEEK

DASHBOARD

Showing July 19 - July 25, 2026

SUN	10-Jul	DEMO FLEET			
DATE	DAY	DEMO SUV	DEMO CROSSOVER	DEMO COMPACT	DEMO PICKUP
		Plate: DEMO-01	Plate: DEMO-02	Plate: DEMO-03	Plate: DEMO-04
		1001001	1001002	1001003	1001004
SUN	7/19		Shop - Central Garage		
MON	7/20		Shop - Central Garage	8:00 AM-5:00 PM Cameron Davis Project coordination	
TUE	7/21		Shop - Central Garage		
WED	7/22		Shop - Central Garage		
THU	7/23		Shop - Central Garage		
FRI	7/24		Shop - Central Garage		
SAT	7/25		Shop - Central Garage		

Weekly schedule: timed bookings and dated unavailability.

Vehicles	Jul-26	Demo SUV (1001001)		Demo Crossover (1001002)		Demo Compact (1001003)		Demo Pickup (1001004)	
Day	Date	Trip(s)	End Mileage	Trip(s)	End Mileage	Trip(s)	End Mileage	Trip(s)	End Mileage
Wednesday	7/1/2026			1	41,825				
Thursday	7/2/2026	1	52,125						
Friday	7/3/2026					1	28,410		
Saturday	7/4/2026								
Sunday	7/5/2026								
Monday	7/6/2026	1	52,210						
Tuesday	7/7/2026							1	62,920
Wednesday	7/8/2026					1	28,520		
Thursday	7/9/2026			1	41,925				
Friday	7/10/2026	1	52,340			1	28,630		
Saturday	7/11/2026								
Sunday	7/12/2026								
Monday	7/13/2026							1	63,020
Tuesday	7/14/2026								
Wednesday	7/15/2026	1	52,420						
Thursday	7/16/2026							1	63,110
Friday	7/17/2026								
Saturday	7/18/2026			IN SHOP	Central Garage				
Sunday	7/19/2026			IN SHOP	Central Garage				
Monday	7/20/2026	1	52,530	IN SHOP	Central Garage	1	28,740		
Tuesday	7/21/2026			IN SHOP	Central Garage				
Wednesday	7/22/2026			IN SHOP	Central Garage				
Thursday	7/23/2026			IN SHOP	Central Garage				
Friday	7/24/2026			IN SHOP	Central Garage				
Saturday	7/25/2026			IN SHOP	Central Garage				
Sunday	7/26/2026			IN SHOP	Central Garage				
Monday	7/27/2026			IN SHOP	Central Garage				
Tuesday	7/28/2026			IN SHOP	Central Garage				
Wednesday	7/29/2026			IN SHOP	Central Garage				
Thursday	7/30/2026			IN SHOP	Central Garage				
Friday	7/31/2026			IN SHOP	Central Garage				
July 2026 TOTALS:		5	530	2	225	4	440	3	310
CUMULATIVE MILEAGES:		Ending Mileage	52,530	Ending Mileage	41,925	Ending Mileage	28,740	Ending Mileage	63,110

Monthly history: trips, ending odometers, mileage, and status.

- RESERVATION LOGIC

 - Single-day and inclusive multi-day bookings; one associated row per vehicle-date.
 - Multiple same-day bookings are allowed when time intervals do not overlap.
 - Half-open [start, end) boundaries permit adjacent bookings; actual overlaps are blocked.
- CONTROLLED CORRECTIONS

 - Recent mileage and reservation records load back into dashboard controls.
 - Grouped multi-day edits expand, contract, or move ranges after conflict revalidation.
 - Grouped deletion removes all associated dates; mileage changes recalculate retained odometer history.
- MAINTENANCE + REPORTING

 - Dated maintenance intervals block affected reservations and preserve history after return to service.
 - Outputs include weekly schedules, monthly mileage, trip counts, scheduled hours, fiscal-year totals, and status.
 - Targeted refreshes update only the outputs affected by a transaction.

RELEASE VERIFICATION

Regression coverage includes startup and navigation; mileage save/edit/delete and odometer recalculation; single-day, multi-day, grouped correction, and boundary reservations; maintenance conflicts and history; report synchronization; formula-error scanning; package integrity; and restoration of Excel events, calculation, and screen updating on success and error paths.

PRIVACY

All shown vehicles, people, identifiers, mileage, purposes, locations, and maintenance events are synthetic. No organization data, branding, confidential records, or external data connections are included.

REVIEW THE IMPLEMENTATION

<https://github.com/cbmons4/vehicle-mileage-reservation-tracker>

The repository includes the macro-enabled demonstration workbook, complete plain-text VBA export, architecture notes, test coverage, and these sanitized screenshots. No measured business time or cost savings are claimed.